

3. Solidity programming knowledge

Smart contract development in the Tron blockchain:

For developing and deploying smart contracts Tron Studio source code needs to be downloaded from <https://developers.tron.network/docs/download-the-code#section-tron-studio>. You can follow the steps to compile the source code from <https://developers.tron.network/docs/tron-studio-intro>.

Here is sample code for a smart contract, developed using Solidity:

```
pragma solidity ^0.4.23;

contract Migrations {
    address public owner;
    uint public last_completed_migration;

    constructor() public {
        owner = msg.sender;
    }

    modifier restricted() {
        if (msg.sender == owner) _;
    }

    function setCompleted(uint completed) public restricted
    {
        last_completed_migration = completed;
    }

    function upgrade(address new_address) public restricted
    {
        Migrations upgraded = Migrations(new_address);
        upgraded.setCompleted(last_completed_migration);
    }
}
```

The VET blockchain

The VET blockchain of VeChain is developed by the VE Foundation, Singapore. According to the official website, “VeChain aims to connect blockchain technology to the real world by providing a comprehensive governance structure, a robust economic model as well as advanced IoT (Internet of Things) integration, and be the pioneers in real-world applications.” VET is focused on real world use cases such as supply chains, medicine and healthcare, luxury and fashion, automobiles, etc. Until the launch of VeChain Thor, VeChain was dependent on Ethereum technology for smart contract development. The following factors differentiate it from other blockchain technologies:

- Built-in governance tools

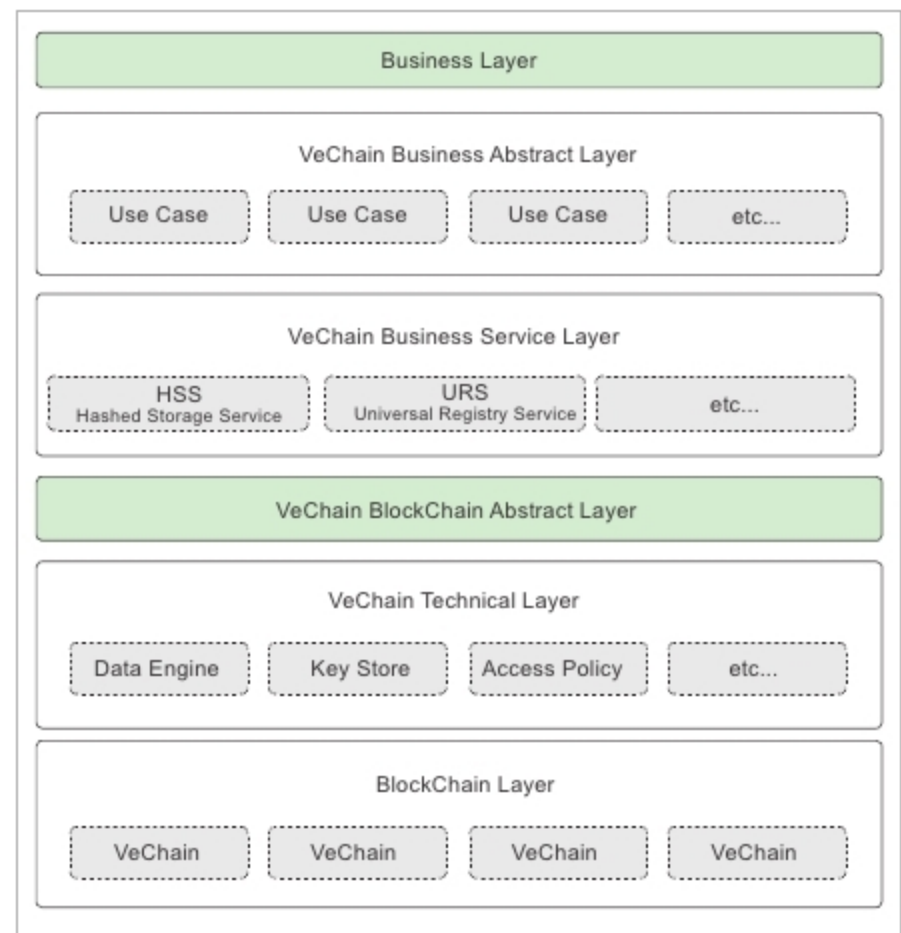


Figure 2: VeChain architecture

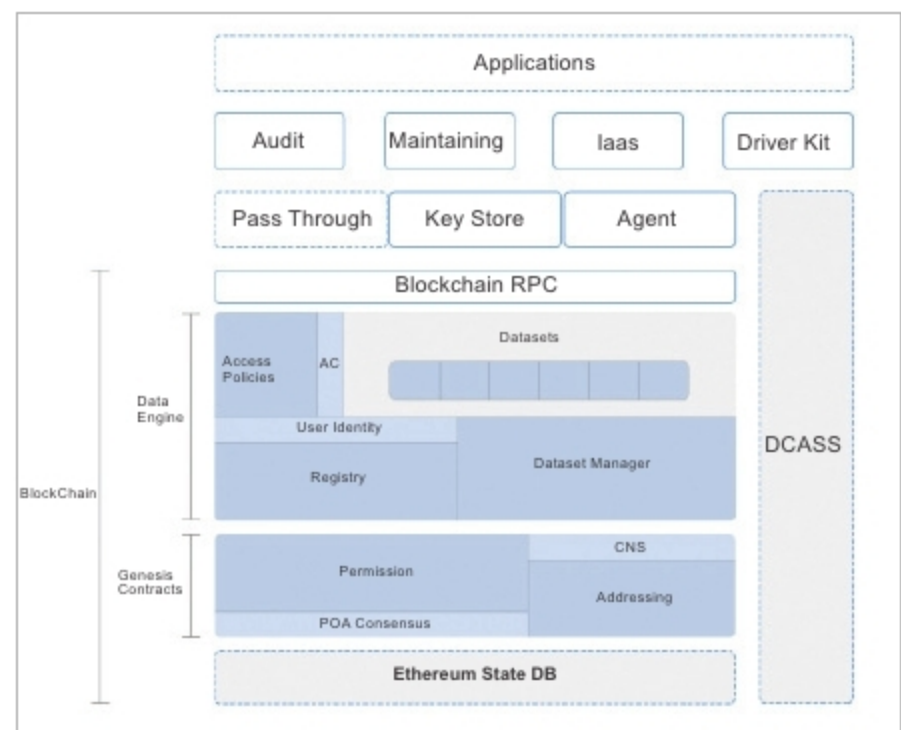


Figure 3: VET blockchain architecture (Image source: *VeChain*)

- Proof-of-authority
- Cryptocurrency disaster recovery plan
- High scalability and sidechain capability

VET uses a combination of blockchain technology and the Internet of Things (IoT) for tracking and tracing items throughout their life cycles. Each item is assigned unique identifiers (RFID tags) using IoT. Thus, each item is logged onto the VeChain blockchain, enabling each stakeholder to track the item.

The NEO blockchain

NEO is the first blockchain to be backed by the Chinese government. By using a digital identity, it aims to digitise